Mirror symmetry for ADE resolutions

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Enumerative Geometry and Geometric Representation Theory, Pisa
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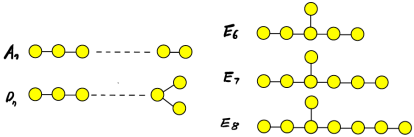
ADE classification

The ADE classification appears both in geometry and representation theory. Given one of the following diagrams:



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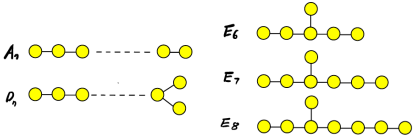


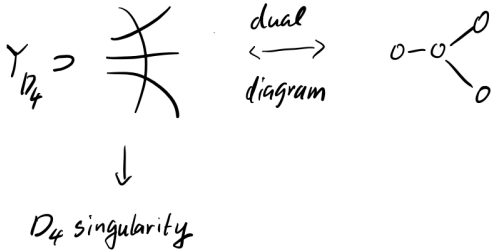
one can associate a

- Isolated surface singularity $X = Z(f) \subset \mathbb{C}^3$: $f = x^2 + y^2 + z^{n+1}$ for A_n.
- Semisimple Lie algebras: sl_{n+1} for A_n.

ADE resolutions

We denote one of the diagram by R, and let X_R and $\mathfrak{g}_R$ be the corresponding singular surface and Lie algebra.

The *ADE* resolutions Y_R are the minimal resolutions of *ADE* singularities X_R.



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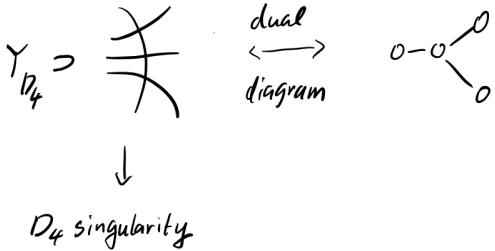

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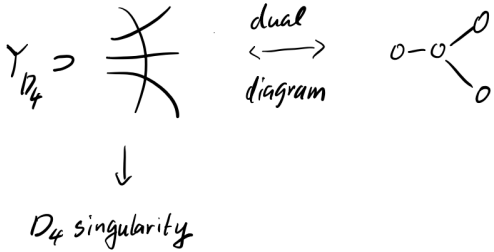

Fact1: $H_2(Y_R;\mathbb{Z}) \cong \Lambda_R$, the later being the corresponding root lattice.

Fact2: $H^*(Y_R;\mathbb{C}) \cong \mathbb{C} \times \mathfrak{h}_R$, the later being the Cartan subalgebra of $\mathfrak{g}_R$.

Quantum connection of ADE resolutions

The Dubrovin connection (or quantum connection) is a $z-$pencil of affine connections on the trivial bundle $H^*(Y) \times U \mapsto U$, where $U \subset H^*(Y)$:

$$\nabla^{(z)}_{\partial_i} X = \nabla_{\partial_i} X + \frac{1}{z} \phi_i \star_\tau X,$$

here $\star_\tau, \tau \in U$ is the quantum product, defined using Gromov-Witten invariants of Y.

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Remark. By Bryan-Gholampour, the quantum product $\star_\tau$ for Y_R can be expressed as a sum over positive roots of $\mathfrak{g}_R$.

Mirror curve from Lie algebras

A family of curve $\Sigma_\tau = \overline{\{P(\tau, \mu, \lambda) = 0\}} \subset \mathbb{P}^2$ on $H^*(Y_R; \mathbb{C}) \cong \mathbb{C} \times \mathfrak{h}_R$ can be constructed from the corresponding semisimple Lie algebra $\mathfrak{g}_R$.

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The equation $P(\tau,\mu,\lambda)$ of the curve is given by the twisted characteristic polynomial of an irreducible representation ρ:

$$P(\tau,\mu,\lambda) = Q(W_i \mapsto W_i + \delta_{i,\bar{k}}\frac{\lambda}{w_0}),$$
$$Q = Det_\rho(e^\tau - \mu\, id) \in \mathbb{Z}[W_1,\cdots,W_n,\mu], W_i = Tr_{\rho_i}(e^\tau),$$

and ρ_i are fundamental representations.

Example: $R = A_1$

The A_1 Lie algebra is $sl_2(\mathbb{C})$, we identify $H^2(Y_{A_1}) \cong \mathbb{C}\{\alpha\}$ where α is the simple root. Write $\tau = x_0 + x_1\alpha \in H^*(Y_{A_1})$.

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Take ρ as the defining representation of $sl_2(\mathbb{C})$, then the characteristic polynomial is

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Moreover, solving λ gives

$$\lambda = e^{x_0}\frac{\mu^2 - (e^{x_1} + e^{-x_1})\mu + 1}{\mu},$$

this is a $\tau-$ family of Hurwitz cover of genus 0.

Mirror symmetry for ADE resolutions

Recently, we were able to prove the following:

Theorem (M '25)

$\int_\Gamma \lambda^{1/z}\phi, \Gamma_\tau \in H_1(\Sigma_\tau, (\lambda))$ *give flat coordinates of quantum connection, for type A and type D resolutions.*

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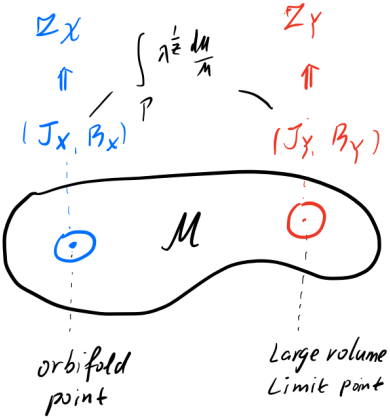
Remark: The logarithm of spectral parameter $\lambda(\tau, \mu) : \Sigma_\tau \to \mathbb{P}^1$ can serve as a super-potential for Y_R.

Crepant resolution conjecture: a work in progress

Y_R is **crepant resolution** of X_R, conjecturely, they give the same information of Gromov Witten invariants. Givental-Teleman reconstruction provides a framework to study this conjecture.



Thanks!